11. What is the activation energy for a reaction if its rate doubles when the temperature is raised from 200K to 400K? ($R = 8.314\ JK^{-1}mol^{-1}$)

a) 234.65 kJ $mol^{-1}K^{-1}$

b) 434.65 kJ $mol^{-1}K^{-1}$

c) 434.65 J $mol^{-1}K^{-1}$

d) 334.65 J $mol^{-1}K^{-1}$

 ; This reaction follows first order kinetics. The rate constant at particular temperature is 2.303×10^{-2} $hour^{-1}$. The initial concentration of cyclopropane is 0.25 M. What will be the concentration of cyclopropane after 1806 minutes? $(\log 2 = 0.3010)$

a) 0.125M b) 0.215M

c) 0.25×2.303M d) 0.05M

12. For a first order reaction, the rate constant is 6.909 min^{-1}.the time taken for 75% conversion in minutes is

a) $\left(\frac{3}{2}\right)\log 2$ b) $\left(\frac{2}{3}\right)\log 2$

c) $\left(\frac{3}{2}\right)\log\left(\frac{3}{4}\right)$ d) $\left(\frac{2}{3}\right)\log\left(\frac{4}{3}\right)$

13. In a first order reaction $x \longrightarrow y$; if k is the rate constant and the initial concentration of the reactant x is 0.1M, then, the half life is

a) $\left(\frac{\log 2}{k}\right)$ b) $\left(\frac{0.693}{(0.1)\,k}\right)$

c) $\left(\frac{\ln 2}{k}\right)$ d) none of these

14. Predict the rate law of the following reaction based on the data given below

$2A + B \longrightarrow C + 3D$

Reaction number	[A] (min)	[B] (min)	Initial rate ($M\ s^{-1}$)
1	0.1	0.1	x
2	0.2	0.1	$2x$
3	0.1	0.2	$4x$
4	0.2	0.2	$8x$

a) $\text{rate} = k[A]^2[B]$ b) $\text{rate} = k[A][B]^2$

c) $\text{rate} = k[A][B]$

d) $\text{rate} = k[A]^{1/2}[B]^{3/2}$

15. Assertion: rate of reaction doubles when the concentration of the reactant is doubles if it is a first order reaction.

Reason: rate constant also doubles

a) Both assertion and reason are true and reason is the correct explanation of assertion.

b) Both assertion and reason are true but reason is not the correct explanation of assertion.

c) Assertion is true but reason is false.

d) Both assertion and reason are false.

16. The rate constant of a reaction is 5.8×10^{-2} s^{-1}. The order of the reaction is

a) First order b) zero order

c) Second order d) Third order

17. For the reaction $N_2O_5(g) \longrightarrow 2NO_2(g) + \frac{1}{2}O_2(g)$, the value of rate of disappearance of N_2O_5 is given as 6.5×10^{-2} mol $L^{-1}s^{-1}$. The rate of formation of NO_2 and O_2 is given

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